Gate B v2 and metrics: change log

What changed, why, what the re-run showed, and what the dashboard now shows.

> **Where this stands.** All six phases are done on the branch gate-b-v2-and-metrics-consolidation, except the push, which you asked to hold. Nothing is merged. Your usual Lightyear folder is unchanged, so this shows on port 8101 only.
>
> **Clenow's latest run:** Gate v2 reads **FAIL net of cost** and **PASS at zero cost**. The only hard block it misses is the bootstrap, at P 0.132 against 0.10. Its factor alpha is red at t 0.66. The result is reported as it fell: nothing was adjusted for it.

Your rulings of 5 November

You said	What I did
Approve, but make them easier: the bar is still very high	Bootstrap P eased from 0.05 to 0.10, and sub-period to 2 of 4 windows, none below -1.0. Pass rate at a true excess Sharpe of 0.5 rose from 42.6% to 57.1%. The alpha bar stays at t 2.0: any lower breaks the 5% false-pass target
Benchmark Sharpe	Set as where the deflated Sharpe bar starts. Dormant while the test is out
Remove the deflated Sharpe for now	Out of the gate, off in config. **So v2 no longer tests for selection luck**. The results table still reports it as a figure
INCOMPLETE is not the right way	Status is PASS or FAIL. A test with nothing to run on shows as not run and is listed beside the status
Good to proceed without those values	AUM, mandate limits, the book and product costs stay blank in config/. Their tests show as not run
Archive, but keep the tearsheet for Excel	**Nothing was archived:** the Excel calculation needs both tearsheet.py and excel_tearsheet.py, and the momentum rotation script is what produces that workbook. All three stay
Hold the push	Not pushed

Commits

Commit	What	Why
925289c	Freeze Gate B v1: version constants, manifest and workbook fields, baseline for every past run	So every later result can be compared with what v1 really said. All 12 runs that had reached a result failed v1.
5139c54	One metrics module: every dashboard, workbook, report and gate reads ros/metrics/core.py	The same metric name gave different numbers depending on where you looked.
fab1b88	Gate v2: hard blocks, evidence scorecard, PASS/FAIL/INCOMPLETE, mandate file, card fields	Built so it could be simulated, before any threshold was chosen.
c471338	Calibrate Gate B by simulation: 8,000 strategies, real-data references, power curves and a proposal	So thresholds come from simulation and not from any one strategy. Clenow was not used.
b26130c	Fixes from the dashboard check: first-day trades count in turnover, older runs label their old Sharpe, runs stop copying core metrics	The dashboard check on the re-run found these.
a73e1e9	Change log for the metrics consolidation and Gate v2 work, with the re-run and dashboard check	You asked for a record of the changes.
fb6bdc7	Approve easier Gate v2 thresholds, deflated Sharpe out, missing inputs show as not run	Your rulings. Committed before any named strategy was scored on these thresholds.
0d5a58f	Wire Gate v2 into Lightyear: scored on every run, new Gate B tab and workbook sheet, past runs re-scored	So every run is scored on v2, on the Gate B tab and in Excel, and past runs can be compared.
6818b7f	Page, workbook and tests name where the cash rate came from	A card naming no rate should not leave Sharpe blank or fail the gate.
0b3ea3f	Reference document updated to v2: one formula per metric, the gate, its calibration, changes from v1; end-to-end gate tests	Phase 6: the document updated to v2.

60 files changed, 4905 insertions(+), 376 deletions(-), compared with the tag gate-v1. Tests: 641 passed, 1 skipped on the final code (570 passed before this work). ruff: no new errors against the original code.

Metric definitions: before and after

Metric	Before	Now	Why
Sharpe	Four formulas: (CAGR - 6%) / vol on the tile, (CAGR - compounded cash) / vol in tables, (mean x 252 - 6%) / vol in Excel, and mean excess / stdev in the deflated Sharpe	One: mean(r - c) / stdev(r - c) x sqrt(252)	One strategy showed several Sharpes under one name
Cash rate	6% typed in the tile and workbook	From the card, or config/fund.yaml if the card names none, and the page says which	No hard-coded rates. 8 of 22 cards name no rate
Max drawdown	Positive in some places, negative in others	Daily values, positive	One sign
Drawdown duration	Calendar days	Trading days	As you specified
Tracking error	Sample in one path, population in another	Sample stdev of daily gaps	One definition
Up / down capture	Daily means in one path, monthly in another	**Monthly means**. My choice: your brief named no frequency	One definition
95% CVaR	Rolling 21-day in one path, daily and negative in another	**Daily, positive loss**. My choice of frequency	As you specified the sign
Turnover	Mean x 252 in one path, sum / years in another	Sum / years	One definition
Bootstrap	Geometric Sharpe, slow loop	The one Sharpe, same settings, plus a paired drawdown gap as evidence only	You kept the method and asked for the one Sharpe

Added

What	Where	Why
Core performance table: the same 16 metrics on every run	Results tab; Core metrics sheet in Excel	Each run's own tables differ. Sortino and Calmar are now on the page
Net of the house cost beside gross	Results tab, Excel. Cost from config/costs.yaml	Your rule: always report gross and net
Gate v2: hard blocks, scorecard, PASS or FAIL	Gate B tab, Gate v2 sheet in Excel. v1 is folded below for comparison	Separates what stops a strategy from what informs you
Capacity	Code done. Shows as not run	Needs AUM in config/fund.yaml
Post-publication metrics	Results tab, Excel, scorecard	Separates the paper's own period from what came after
Price-index label and a total-return hook	Results tab, Excel	A price benchmark flatters relative figures
Diagnostics section	Results tab and Excel Validation sheet	Rows tied to one strategy's rules are not core performance
gate_version and metrics_version	manifest.json, workbook header, Gate B tab	Every result says what produced it
Card fields: n_configs_selected_from, n_sensitivities, rules_fixed_in_print, fits_parameters	ros/cards/schema.py	Needed when the deflated Sharpe returns. fits_parameters already decides walk-forward or sub-period
Config files: costs, fund, mandate, gate thresholds	config/	Settings are read, not typed

Cut and kept

Item	Status
Mean drawdown, vol consistency, hit rate, monthly CVaR, the geometric and duplicate Sharpes	**Cut** from the pipeline report and metrics modules
Sterling, Treynor, Omega, Gain to Pain, tail ratio, adjusted beta, Jensen's alpha, extreme capture, Ret +Ve/-Ve, positive/negative volatility, average annual drawdown, 99% VaR/CVaR	**Off Lightyear, still in the tearsheet.** They stay because you need the Excel calculation. Its numbers can differ from the dashboard's
Geometric Sharpe	**Kept in one place** (ros/metrics/legacy_v1.py): a paper's printed Sharpe compares on its own definition
validate/independent.py	**Kept separate on purpose:** the clean-room cross-check imports nothing from ros
universal_backtester/validation.py	**Not consolidated.** Its own bootstrap and deflated Sharpe remain, but no Lightyear path or gate uses them

Gate v2 and the calibration

Share passing	True Sharpe 0	0.5	0.75	1.0	Factor only	Best of 20	Best of 100
Status PASS (hard blocks)	6.9%	63.2%	87.4%	95.7%	73.4%	72.5%	97.3%
PASS and all green	4.5%	57.1%	84.7%	95.3%	2.6%	59.6%	95.9%

What easing cost. PASS and all green meets the targets for no-skill strategies and factor-only strategies. **Status PASS alone does not:** 73% of factor-only strategies and 6.9% of no-skill ones read PASS, because the scorecard does not block. With the deflated Sharpe out, 96% of the best of 100 random strategies get PASS and all green, against 0% before. So report every version tried, and read the factor alpha row beside the status.

Thresholds were set from simulations and committed (fb6bdc7) before Clenow was scored on them. To be open about it: I had already seen Clenow's gross P of 0.054 and its v1 alpha when I chose them, but not its net P. Formulas and the full curves: financial-metrics-and-gate-b-v2.pdf. gate-b-v2-calibration.pdf is the earlier proposal, with the deflated Sharpe still in.

Clenow, latest run on universe v2: v1 against v2

Test	Gate v1, as run	Gate v2, net of 30bp	Gate v2, as run (zero cost)
Status	**FAIL**, 1 of 5 blocking	**FAIL**	**PASS**
Edge over the benchmark	Not tested	5.22%: PASS	7.05%: PASS
Paired bootstrap P	0.054: fail at 0.05	0.132: FAIL at 0.10	0.054: PASS at 0.10
Worst window	-0.24 absolute Sharpe: fail at 0.30	Sharpe gap 0.02, 4 of 4 windows ahead: green	Sharpe gap 0.11, 4 of 4 windows ahead: green
Factor alpha t	1.52 on 2 factors: fail at 2.0	0.66 on 7 factors: red	1.17 on 7 factors: amber
Deflated Sharpe	0.00: fail	Out of the gate	Out of the gate
After publication: edge	Not tested	3.75%: green	5.51%: green
Data integrity	Not tested	PASS	PASS
Mandate, capacity, book	Mandate passed on the card's flag; no book	Not run: no values supplied	Not run: no values supplied

Run 20261105-132328-ab35. Its daily returns, trades and holdings are byte-identical to 20261102-105054-0427. Its manifest says gate_version v1 because the script ran before the gate moved; the workbook and tab show v2 after Rebuild.

On the page now	As run (zero cost)	Net at 30bp	NIFTY 500
CAGR	19.80%	17.95%	12.83%
Sharpe	0.83	0.74	0.49
Sortino	1.10	0.97	0.65
Calmar	0.88	0.75	0.34
Maximum drawdown	22.47%	23.99%	38.30%
Drawdown duration (trading days)	547	699	490
Information ratio	0.49	0.36	–

Every past run: v1 against v2

All 13 runs failed v1. Under v2, net of the house cost, 6 pass and 7 fail. Five of the passes are the same earlier Clenow run on the older universe. Full detail: docs/gate_b/v1_vs_v2.csv.

Run	Strategy	v1	v2	Net edge	P (net)	Why it changed
20261006-181257-0394	Padhy 130/30	FAIL	**FAIL**	5.59%	0.484	Still fails: bootstrap P above 0.10
20261006-lee-swaminathan-1998	Lee-Swaminathan	FAIL	**FAIL**	-11.79%	0.941	Still fails: bootstrap P above 0.10
20261013-161034-6649	1_s2_0 (s0970389623000538)	FAIL	**PASS**	6.56%	0.031	Passes every hard block that could be run. Alpha no longer blocks (t 0.47), and the deflated Sharpe is out
20261016-102923-bdc1	Clenow	FAIL	**FAIL**	7.15%	0.254	Still fails: bootstrap P above 0.10
20261021-104144-801e	Clenow	FAIL	**FAIL**	7.15%	0.254	Still fails: bootstrap P above 0.10
20261022-153434-39aa	Momentum 50 rebuild	FAIL	**FAIL**	4.25%	0.312	Still fails: bootstrap P above 0.10

Run	Strategy	v1	v2	Net edge	P (net)	Why it changed
20261030-124039-5511	Clenow	FAIL	**PASS**	6.33%	0.070	Passes every hard block that could be run. Alpha no longer blocks (t 1.03), and the deflated Sharpe is out
20261030-171327-0aa8	Clenow	FAIL	**PASS**	6.33%	0.070	Passes every hard block that could be run. Alpha no longer blocks (t 1.03), and the deflated Sharpe is out
20261030-173539-8fd8	Clenow	FAIL	**PASS**	6.33%	0.070	Passes every hard block that could be run. Alpha no longer blocks (t 1.03), and the deflated Sharpe is out
20261030-183103-3639	Clenow	FAIL	**PASS**	6.33%	0.070	Passes every hard block that could be run. Alpha no longer blocks (t 1.03), and the deflated Sharpe is out
20261030-191604-3399	Clenow	FAIL	**PASS**	6.33%	0.070	Passes every hard block that could be run. Alpha no longer blocks (t 1.03), and the deflated Sharpe is out
20261102-105054-0427	Clenow	FAIL	**FAIL**	5.22%	0.132	Still fails: bootstrap P above 0.10
20261105-132328-ab35	Clenow	FAIL	**FAIL**	5.22%	0.132	Still fails: bootstrap P above 0.10

Dashboard check

I opened the re-run in a real browser on the final code. **38 of 38** checks on the metrics and **19 of 19** on Gate v2 pass, with no script errors. Excel, after recalculation, matches the page on CAGR, volatility, Sharpe, drawdown, Sortino and Calmar.

Problems found and fixed

Problem	Cause	Fix
Turnover read 10.33 and 10.26. Net figures slightly too high	**My bug.** The first day's purchases were dropped	Counted on the first day. Net CAGR 17.95% now matches the full re-simulation
Older runs said "Sharpe" but showed the old 6% number	My relabel	It reads "Sharpe (old 6% hurdle)" and says to press Rebuild
Two past runs read FAIL with no test values	Their cards name no cash rate, as do 8 of 22 cards	A fallback rate in config/fund.yaml, named on the page. **Please check 6% is right**
7 rule-specific rows sat in Costs & execution	Written during the run	Lightyear now moves them to Diagnostics, on the page and in Excel
A factor equal to the strategy gave alpha t 17.6	A series regressed on itself	An identical factor is left out, and the row says so

Still open

Item	Why it matters
Mandate, capacity and book tests show not run	A PASS has not been tested on these. Filling in config/fund.yaml and config/mandate.yaml switches them on with no code change
The Gate B brief and "Evidence supports" are written from the v1 checklist	They can disagree with the v2 status above them. Labelled on the page. Worth moving the brief to v2
Each run can still add its own rows	"Rolling one-year behind" shows 34.59% and 34.78% on this run. Runs are told not to, but nothing enforces it
Older runs show v2 only after Rebuild	The table above scores them all, but their pages need Rebuild
Your usual Lightyear on port 8100	It is on the old branch, and was not running when I last checked. Merging is your call